A Self-Supervised Spaceborne Multispectral and Hyperspectral Image Fusion Unrolling Network

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***Abstract*— Deep learning has emerged as the predominant approach for multispectral and hyperspectral image fusion. However, most fusion networks are typically trained and validated on hyperspectral and multispectral image pairs generated from the same hyperspectral images, with degradation simulations inconsistent with real situations and relatively limited volumes of images. When transferring a pretrained multispectral and hyperspectral image fusion model from ground or airborne images to spaceborne images, it encounters a larger dataset and more complex spatial-spectral degradation, leading to spectral distortions and spatial artifacts in the fused images. In this article, the challenges associated with the transfer are addressed through the introduction of a self-supervised multispectral and hyperspectral image fusion unrolling network for spaceborne imagery, termed as MH-FUNet. MH-FUNet adopts a self-supervised paradigm to learn a robust mapping from spaceborne data. It utilizes a deep unrolling network to iteratively refine fusion results from coarse to fine. To account for spatial scale differences between the self-supervised training and test datasets, a multiscale fusion strategy is introduced. This strategy is combined with spectral and spatial attention mechanisms to restore spatial and spectral details. Additionally, a gradient constraint unit is proposed to maintain spatial consistency when up-scaling low-resolution hyperspectral imagery. Performance evaluation of the proposed method is conducted against state-of-the-art fusion techniques on both simulated Chikusei dataset and the proposed real WHU-MHF dataset, which consists of simultaneously observed hyperspectral and multispectral image pairs. MH-FUNet outperforms existing methods across all datasets, demonstrating superior performance in spaceborne multispectral and hyperspectral image fusion experiments.**

***Index Terms*— Deep unrolling, gradient constraint, multiscale fusion, multispectral and hyperspectral image fusion, spaceborne.**

I. Introduction

DUE to the limited energy, there always exists trade-offs among spatial resolution, spectral resolution, and signal-to-noise ratio (SNR). As a result, the spaceborne hyperspectral imager are with low spatial resolution (LR, i.e., 30 m). To make up the LR of hyperspectral imager for remote sensing interpretation applications [1], [2], [3], it is common to equip satellites with a high spatial resolution (HR) panchromatic or multispectral imager as a supplement (i.e., EO-1, ZY1-02D) [4], [5]. To combine the information of HR multispectral imagery with the information of LR hyperspectral imagery, multispectral and hyperspectral image fusion have attracted considerable attention in recent years [6], [7].

Generally, multispectral and hyperspectral image fusion methods can be briefly divided into the following categories.

1) Pan-sharpening based fusion methods, including two kinds of methods, the component substitution (CS)-based methods [8], [9], [10], [11], [12], and the multiresolution analysis [13] (MRA)-based methods [14], [15]. CS-based methods involve substituting the intensity of the LR data with the one of HR data, while MRA-based methods aim to incorporate the fine frequential feature from the HR data into the LR data. Though easy to apply, these methods tend to introduce noticeable spatial-spectral distortion.
2) Variational-based methods, which model the fusion problem as an ill-conditioned inverse process corresponding to the observation model. To solve the problem, iterative optimization is introduced to solve the problem with the use of certain prior knowledge, such as a total variation prior, sparse prior, low-rank prior [16], [17], [18], [19], [20], [21], [22], [23], [24], [25], [26], [27], [28]. However, they often depend on a manually designed regularization. If the design of the prior is too simple, the performance will be unsatisfactory. Otherwise, the model would be too hard to optimize.
3) Deep learning methods, which have demonstrated its effectiveness in capturing complex and nonlinear relationships within data, have been extensively applicated to various low-level vision domains. Over the past few years, there has been a surge in the number of deep-learning-based multispectral and hyperspectral image fusion methods proposed [29], [30], [31], [32], [33], [34], [35], [36], [37], [38], [39], [40]. Deep-learning-based methods employ a nonlinear mapping approach to fuse LR hyperspectral and HR multispectral data into HR hyperspectral imagery. Following the Wald's protocol [41], the network learns the mapping by training with simulated image pairs on lower resolution. Compared with the traditional methods, deep-learning-based

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(a) (b) (c)

Fig. 1. Visual comparison between the fusion results with, without spatial consistency and GT. Artifacts could be alleviated in the proposed network. (a) GT. (b) Fusion without spatial consistency. (c) Fusion with spatial consistency (ours).

methods require little prior knowledge, and can obtain superior fusion results.

When applying current deep learning methods to the task of spaceborne image fusion, problems arise on both the dataset and the fusion strategy.

1) The Wald's protocol works well on high-quality airborne hyperspectral images, but it poses a new challenge for spaceborne hyperspectral-multispectral image pairs [42]. For example, in the referred experiment, SPOT images were used for image fusion task, with 10–20 m resolution, respectively. Then the resolution for the degraded image was 20–40 m. However, satellite hyperspectral images commonly have a 30 m resolution, and are degraded to 90 m for training. Severe detail loss occurs due to the degradation. Therefore, the fusion task is easier for the network to learn on this resolution, which may cause insufficient detail recovery back on the original one. In such a case, these methods failed to meet the Wald's protocol on space-borne cases.
2) Current methods often overlook the spatial differences between the image pairs. Though systematic geometric calibration was conducted, the regional differences in detail still existed. Direct fusion of the image pairs would lead to severe artifacts in crucial spatial details such as small targets, shapes, and edges, as illustrated in Fig. 1. This phenomenon could also be observed in certain cases [43], [44], [45]. The common idea of the effected methods is that they adopt the idea of spectral super-resolution. Therefore, for image pairs with significant differences in detail, the wrong mapping of spectral information will manifest as abnormal spots in space. In the following experiment, this phenomenon could also be observed. In this case, full consideration should be given to correctly restoring the correct structural information in the fused images, which is referred to as *spatial consistency* in the following context.

In this article, to overcome the problems faced in the multispectral and hyperspectral image fusion tasks, innovative strategies are introduced to the data, training strategy, and network, respectively. To construct an effective fusion strategy, a deep self-supervised unrolling network (MH-FUNet) for spaceborne multispectral and hyperspectral image fusion is proposed, aiming to solve the problems of spatial inconsistent in train-test data and scale-variance. For the utilization of data, a benchmark dataset for spaceborne multispectral and hyperspectral image fusion is introduced, which is named the Wuhan University Multispectral-Hyperspectral Fusion (WHU-MHF) dataset. This article's primary contributions can be summarized as follows.

1) The proposed unrolling network (MH-FUNet) for multispectral and hyperspectral image fusion. Within the unrolling architecture, the forward degradation process can be easily modeled as network components and learned through data-driven approaches. In this way, degradation modeling can be well integrated with a network structure, guiding the network to explicitly learn useful content.
2) A gradient constraint unit is introduced to constrain the feature maps extracted separately from the LR hyperspectral and HR multispectral data. Applying the spatial constraint from the image gradient domain, the spatial features can achieve an image motion compensation effect.
3) A standard benchmark spaceborne multispectral and hyperspectral image fusion dataset (WHU-MHF), which is composed of image pairs obtained simultaneously by the multispectral and hyperspectral sensors carried on ZY-1-02D. The dataset covers water, buildings, vegetation, and other important features.

II. Related Work

A. Multispectral and Hyperspectral Image Fusion Model

The notations used in this article are summarized as follows: $\mathbf{X} \in \mathbb{R}^{M\times N\times S}$ is used to refer to the target hyperspectral image, with $M \times N$ pixels in each of the S bands. $\mathbf{Y} \in \mathbb{R}^{m\times n\times S}$ denotes the observed hyperspectral image with $m \times n$ pixels, and $\mathbf{Z} \in \mathbb{R}^{M\times N\times s}$ for the observed multispectral image of s bands. $\lambda = M/m = N/n$ is the spatial resolution ratio between the observed HR multispectral image and the LR hyperspectral image.

For remote sensing observation, as mentioned before, the HR hyperspectral image cannot be directly acquired. Instead, the observation of the HR multispectral image and LR hyperspectral image, which is the degradation process, can be described by the following mathematical formulation:

$$[\mathbf{Y}, \mathbf{Z}] = g(\mathbf{X}) \tag{1}$$

where function g denotes the spatial and spectral degradation process from HR hyperspectral images to LR hyperspectral and HR multispectral image pairs, which can be expressed as

$$\begin{aligned} \mathbf{Y} &= \mathbf{XB} \\ \mathbf{Z} &= \mathbf{RX} \end{aligned} \tag{2}$$

where $\mathbf{B}$ represents the spatial blurring and down-sampling process, resulting in the loss of spatial details, and $\mathbf{R}$ represents the spectral sensor integration, which leads to the band reduction. The recovery of the desired HR hyperspectral image involves the formulation of the fusion process as the inverse process corresponding to the image degradation

$$\mathbf{X} = f(\mathbf{Y}, \mathbf{Z}) \tag{3}$$

where the function f stands for the fusion process. Hence, the objective of multispectral and hyperspectral image fusion is to

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address the problem posed by (1) and (3)

$$\widehat{\mathbf{X}} = \arg\min_{\widehat{\mathbf{X}}} \| f\left(g\left(\widehat{\mathbf{X}}\right)\right) - \mathbf{X}\|. \tag{4}$$

Here, the data self-consistency is employed, since there is no HR hyperspectral image at the desired resolution, it is necessary to employ the training process on the lower-resolution image.

B. Multispectral and Hyperspectral Image Fusion Methods Based on a CNN

Deep learning has made remarkable strides in terms of efficiency and accuracy, resulting in its extensive adoption in various low-level image-processing tasks. Convolutional neural network (CNN)-based methods are particularly popular due to their local translation invariance and rotation invariance, enabling the utilization in the hyperspectral and multispectral image fusion tasks. These deep learning-based fusion approaches aim to learn a nonlinear mapping function f, where the network parameters Θ are learned through an optimization process, as shown in the following equation:

$$\mathbf{X} = f(\mathbf{Y}, \mathbf{Z}, \Theta) \tag{5}$$

$$\Theta = \arg\min_{\Theta} \| f(\mathbf{Y}, \mathbf{Z}, \Theta) - \mathbf{X}\|. \tag{6}$$

Early deep learning algorithms can be divided into single branch networks and dual branch networks. The idea behind the single branch methods was to extract the high-resolution features and the low-resolution features with one identical network. For example, the PanNet [29] obtained the fused image through the stacked features extracted from up-sampled observed hyperspectral image with the corresponding multispectral image. Dong et al. [30] introduced the super-resolution CNN (SRCNN), which was a deep CNN for single-image super-resolution that can achieve a superior performance. In SSRNET [46], a single branch network was introduced to model the correlations both in the spatial and spectral domain. In [47] the 3DCNN fusion was proposed, which utilized 3-D CNNs that simultaneously learned the spatial and spectral information by inputting the feature image after principal component replacement. However, these methods left out the scale difference between the image pairs to be fused, which was hard to be fully extracted from one network.

Dual-branch CNN-based fusion approaches employ subnetworks to extract features from LR hyperspectral images and HR multispectral images, combining them to predict an HR hyperspectral image [34]. Wang et al. [36] introduced the deep blind iterative fusion network (DBIN), a CNN-based fusion method utilizing two paths to extract spatial and spectral features from the hyperspectral and multispectral images. In [48], an attention mechanism was introduced into the network, which aims at improving the detail preserving. In [49], edge preserving was achieved by utilizing the spatial-frequential feature extraction. The dual branch network structure effectively distinguishes features from different images, but how to design efficient feature extractors and joint multiscale features remains to be studied.

C. Unrolling Algorithms for Multispectral and Hyperspectral Image Fusion

The deep unrolling technique, which is also known as deep unfolding, is a connection between knowledge and data [50]. More precisely, for an inverse problem in image quality recovery, there is an intrinsic observation model that describes the degradation model from an ideal high-quality image to the real one, including blurring, noise, etc. Since it is usually difficult to find the analytical solution for the objective function built by the original degradation model, optimization is introduced to iteratively obtain a numerical solution. This is how a regularization method works. Deep unrolling involves unrolling the iterative process into network parameters, which can be learned through training. A deep unrolling algorithm was first proposed by Gregor and LeCun [51] in the context of sparse coding. The first unrolling method designed for the image fusion task was proposed in [47], where the dimensionality reduction of the LR hyperspectral image was performed through principal component analysis. The main components were then input into a 3-D CNN fusion block to be fused with the HR multispectral image. This dimensionality reduction operation was utilized to reduce the computational cost and make the method more robust to noise. In [52], a denoising prior was introduced through a new kind of recursive structure. To be more specific, the original optimization function was split into two parts: one made up of the spatial-spectral consistency, and the other equivalent to an intrinsic denoising problem. Dian et al. [35] propose a fusion method that leverages subspace representation and a pretrained CNN for gray image denoising. The key advantage of this approach was its ability to directly apply the CNN denoiser to hyperspectral and multispectral images without retraining. In [45], a convolutional sparse coding model was proposed, which improves the traditional observation model by separating the features as shared and unique ones. According to the modeling, the different feature extraction procedures were unrolled as different network units. In [53], a supplement model was introduced to make up for the iterative error of the linear unmixing model. These fusion frameworks present a practical and effective solution for a wide range of remote sensing applications, offering promising results and simplifying the fusion process for real-world applications.

III. Deep Self-Supervised Unrolling Network for Multispectral and Hyperspectral Image Fusion

A. Motivation

The data characteristics of spaceborne multispectral and hyperspectral imagery pose significant challenges for fusion tasks. On the one hand, to ensure the accuracy of learned mappings, images are degraded from the original image pairs, rather than from single HR hyperspectral images in simulated scenes. However, due to factors such as observational conditions, there exists spatial inconsistency between images from multiple sensors, which can result in artifacts in the fused imagery. On the other hand, the large-scale difference in spaceborne imagery, such as 30–90 m resolution variations, lead to significant difference in image features. How to enable the

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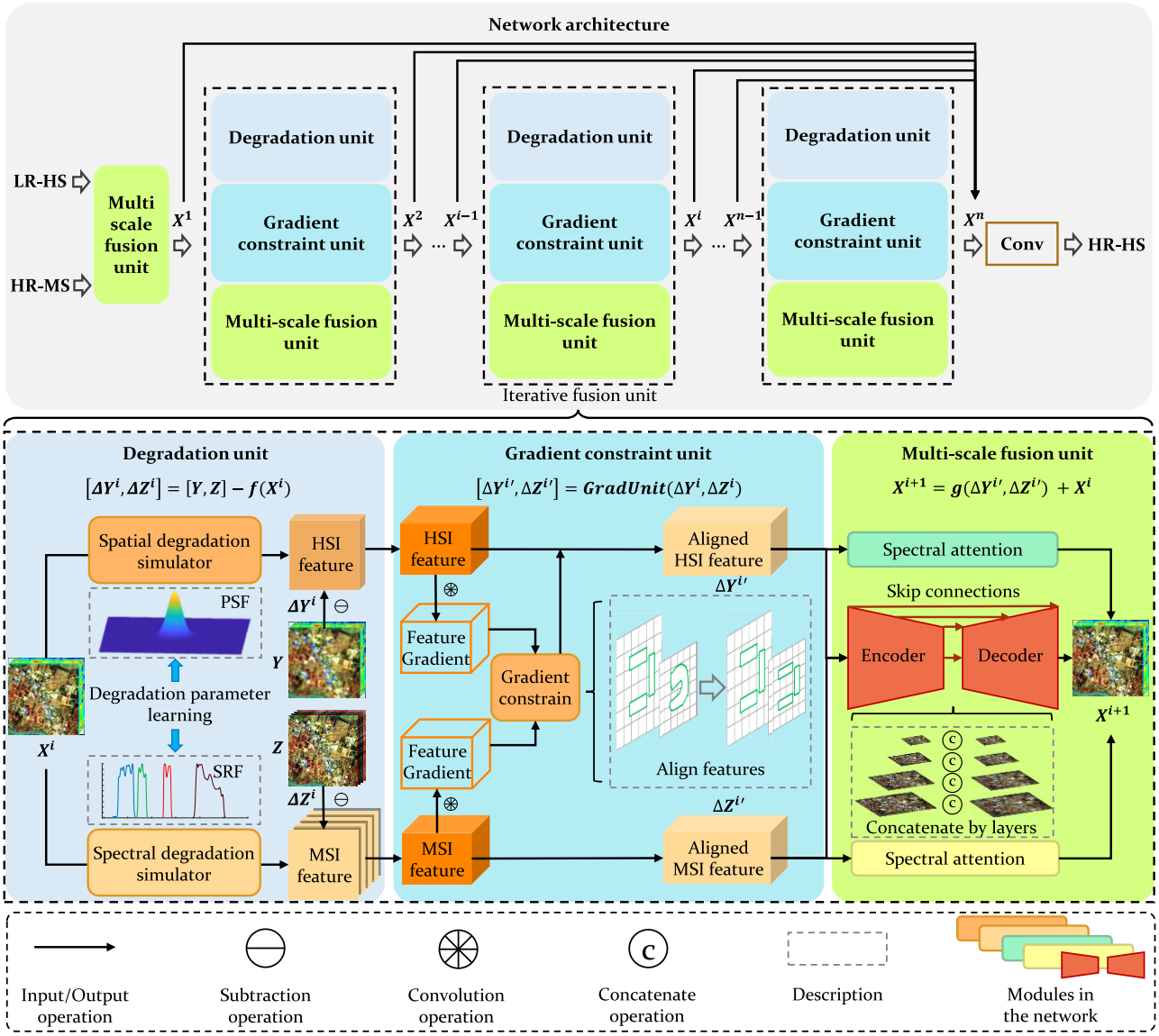


Fig. 2. Flowchart of the proposed MH-FUNet framework is composed of sequential fusion units to iteratively refine the fusion result. Each fusion unit includes a degradation unit, a gradient constraint unit, and a multiscale fusion unit, corresponding to one *degradation-alignment-fusion* iteration. For better fusion performance, fusion results from every iterative fusion unit are concatenated and then convoluted to get the final fusion image.

network to learn better cross-scale features in a self-supervised paradigm is the major challenge.

To address the aforementioned challenges, a framework is designed for robust remote sensing hyperspectral and multispectral image fusion task. Fig. 2 illustrates the architectural design of our proposed framework. First, to effectively utilize both the target fusion data and the degraded input data, a deep self-supervised unrolling network to better learn the fusion results is needed. For robust cross-scale fusion capacity, a multiscale fusion unit is proposed to adapt to the varying spatial scales inherent in spaceborne imagery, ensuring accurate feature integration. For spatial alignment for the features, a gradient constraint unit that enforces spatial consistency during the up-scaling process is introduced to mitigate spatial artifacts.

B. Overall Unrolled Architecture of MH-FUNet

As for the unrolling-based network, the basic idea of this algorithm is that if the desired fused image $\widehat{\mathbf{X}}$ is exactly equal to the HR hyperspectral image, and the process of degrading the HR hyperspectral image to obtain the HR multispectral image and LR hyperspectral image is in line with the actual situation, then the degraded LR hyperspectral image $\widehat{\mathbf{Y}}$ and HR multispectral image $\widehat{\mathbf{Z}}$ should be the same as the original LR hyperspectral image $\mathbf{Y}$ and HR multispectral image $\mathbf{Z}$. If they are different, their errors can be back-projected onto $\widehat{\mathbf{X}}$ through a potential fusion model to correct them. When implementing this algorithm, since the exact fusion model and degradation model are potentially unknown in real cases, a technique referred to as algorithm unrolling is introduced to encode the fusion model and degradation model into parameters in the network, which can be automatically learned through data-driven supervised learning. For the degradation model, the process can be expressed as

$$\begin{aligned} \mathbf{Y} &= f_{\mathrm{PSF}}(\mathbf{X}) \\ \mathbf{Z} &= f_{\mathrm{SRF}}(\mathbf{X}) \end{aligned} \tag{7}$$

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where f_{PSF} denotes the spatial and f_{SRF} denotes the spectral degradation process from HR hyperspectral to LR hyperspectral image and HR multispectral image. These functions are parameterized as the degradation unit in every iterative refine unit. After been fused at one stage, the degradation model is introduced for data consistency at the following stage. For spatial degradation on the LR hyperspectral image, the optical system functions can be viewed as a blurring matrix in the spatial domain. A series of 2D-convolutional layers with a kernel size of $2^*\lambda + 1$ and stride of λ so that the spatial degradation at scale λ could be learned as an approximation of the point spread function (PSF) from the datasets. As for the LR hyperspectral image, the spectral degradation is called the spectral response function (SRF), which describes the linear coefficient in the spectral domain from the HR hyperspectral to HR multispectral imagery. In the network, to force the network to focus more on the spectral information, convolutional layers with a size of 1×1 are introduced to learn the spectral mapping. After the degradation, feature alignment is conducted through the proposed gradient constraint unit.

As for the fusion unit, a multiscale spatial-spectral attention fusion unit is proposed, which can be expressed as

$$f(\mathbf{Y}, \mathbf{Z}) = \mathrm{MS}(\mathbf{Y}, \mathbf{Z}) \otimes \mathrm{CA}(\mathbf{Y}) \odot \mathrm{SA}(\mathbf{Z}) \tag{8}$$

where function f represents the potential fusion model; MS represents the proposed multiscale learning unit; and CA and SA represent the attention mechanism along the spectral dimension and spatial dimensions, respectively. The iterative process of the algorithm is mapped to a deep residual network design, where the kth iterative process is a combination of the image degradation unit and image fusion unit. For the residual learning, the residual error between the degraded images and the original images is fed into the subsequent fusion network structure, instead of the degraded images themselves. These blocks are encoded to a network in a residual learning approach

$$\mathbf{X}^{k+1} = \mathbf{X}^k + f_k\left(\Delta\mathbf{Y}^k, \Delta\mathbf{Z}^k\right) \tag{9}$$

where $\Delta\mathbf{Y}^k$ and $\Delta\mathbf{Z}^k$ are the residual errors between the kth degraded $\mathbf{Y}^k$ and $\mathbf{Z}^k$ and the original LR hyperspectral image and HR multispectral image. $\Delta\mathbf{Y}^k$ and $\Delta\mathbf{Z}^k$ are calculated as follows:

$$\left[\Delta\mathbf{Y}^k, \Delta\mathbf{Z}^k\right] = [\mathbf{Y}, \mathbf{Z}] - \left[\mathbf{Y}^k, \mathbf{Z}^k\right] = [\mathbf{Y}, \mathbf{Z}] - g_k\left(\mathbf{X}^k\right). \tag{10}$$

Subsequently, the final image is generated by comprehensively considering the fusion image generated in the iterative process

$$\widehat{\mathbf{X}} = conv\left(cat\left(\mathbf{X}^1, \ldots, \mathbf{X}^{\mathbf{k}}\right)\right). \tag{11}$$

With the formulation of the above estimation, the direct estimation error between $\widehat{\mathbf{X}}$ and $\mathbf{X}$, which is difficult to precisely measure, is transferred to estimate the error between the image pairs to be fused. This strategy, which is known as *data consistency*, has been proven to be effective in many inverse problems [54]. The entire unrolled algorithm is shown in Algorithm 1.

Algorithm 1 The Proposed Unrolled Algorithm

Input: LR hyperspectral image $\mathbf{Y} \in \mathbb{R}^{m\times n\times S}$, HR multispectral image $\mathbf{Z} \in \mathbb{R}^{M\times N\times s}$, $\mathbf{X}^1 = F(\mathbf{Y}^1, \mathbf{Z}^1)$
for $l = 2, \ldots, k-1$ **do**
Calculate the degraded $\mathbf{Z}^l$ and $\mathbf{Y}^l$ using (2)
Obtain the residual features of $\Delta\mathbf{Z}^l$ and $\Delta\mathbf{Y}^l$ by $\Delta\mathbf{Y}^l = \mathbf{Y} - \mathbf{Y}^l$, $\Delta\mathbf{Z}^l = \mathrm{Z} - \mathbf{Z}^l$
Extract the features and align the features by the following equation: $[\Delta\mathbf{Y}^{l'}, \Delta\mathbf{Z}^{l'}] = GradientUnit(\Delta\mathbf{Y}^l, \Delta\mathbf{Z}^l)$
Fuse the features and reconstruct the desired $\mathbf{X}^{l+1}$ by (7) and (8)
end for
$\widehat{\mathbf{X}} = conv(cat(\mathbf{X}^1, \cdots, \mathbf{X}^{\mathbf{k}}))$
Output: Fused HR hyperspectral image $\widehat{\mathbf{X}}$

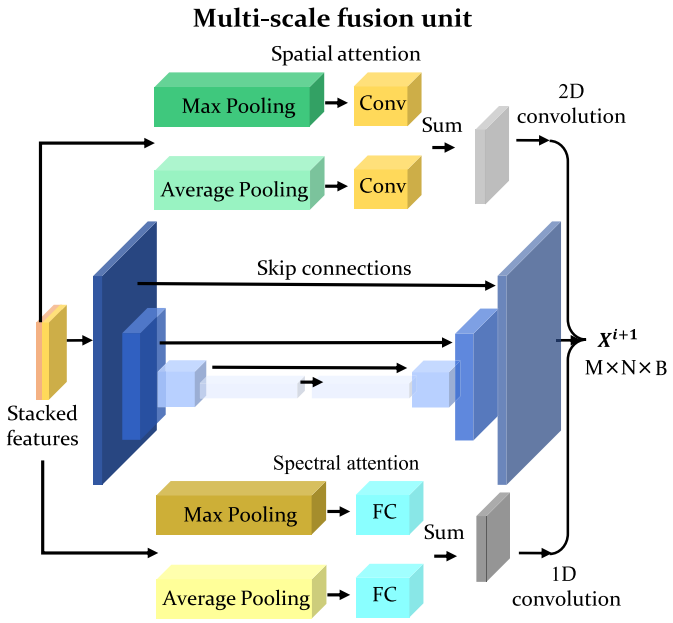


Fig. 3. Architecture of the multiscale fusion unit.

C. Multiscale Fusion Unit

The multiscale fusion unit consists of spatial attention, spectral attention, and the fusion component. First, a residual convolution unit to extract features from the up-sampled LR hyperspectral and HR multispectral images, and then concatenate them. The fusion process is then divided into encoding and decoding parts. The encoding part extracts multiscale image spatial and spectral features through down-sampling and convolution. The decoding part reconstructs the fused image from feature to image by means of transposed convolution and a skip connection, to obtain information from different scales to realize the fusion of the multiscale information. The architecture of the multiscale fusion unit is shown in Fig. 3.

Since there is a resolution difference between the training data and the target data, the network would learn a mapping at the original resolution instead of a relative scale, which would turn out to be blurring when directly applying the image to the original resolution. To reduce this phenomenon, the fusion block is encoded to a multilayer structure. By means of learning the features at different resolutions, the network is forced to learn the mapping from different spatial scales.

First, a residual convolution unit is used to extract features from the upscaled LR hyperspectral image and HR

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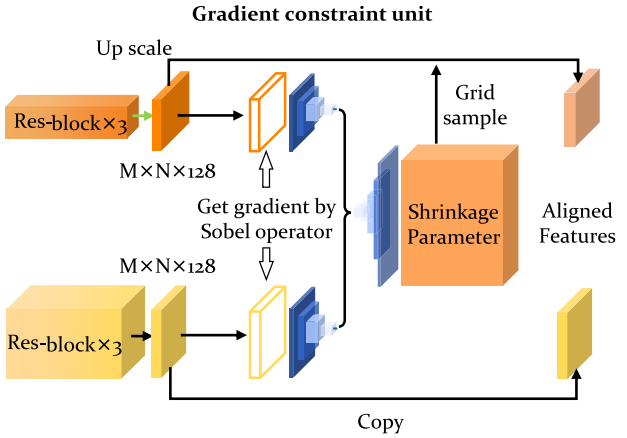


Fig. 4. Architecture of the gradient constraint unit.

multispectral image, and concatenate the two feature maps. The residual convolution block is composed of two convolutional layers and a corresponding rectified linear unit (ReLU) activation function with a batch normalization layer. The size of the convolution kernel used by the convolutional layer is $3 \times 3 \times 128$, and the image edge is filled to maintain the size of the feature map. The unit divides the fusion process into two parts: encoding and decoding. The encoding part extracts multiscale spatial and spectral features of the imagery through down-sampling and convolution. The down-sampling incorporates a $3 \times 3 \times 128$ convolution kernel with a stride of 2 to simulate the complex nonlinear relationship between different scales. The decoding part decodes and reconstructs the fused image from feature to image by means of transposed convolution and skip connection. It obtains information from different scales to realize the fusion of multiscale information and learn the differences and common points between the different scales of information, so as to get rid of the dependence on the original scale, and allow the fusion model to be suitable for different scales.

The attention mechanism can recalibrate the mapping of features. It is a very popular network structure in the computer vision field. The spatial attention mechanism can give weights based on features, which essentially achieves the effect of transforming the spatial information to another space for preservation, and can make better use of the spatial detail information of the multispectral imagery.

D. Gradient Constraint Unit

Simply using an encoder–decoder structure is not enough for spatial inconsistency that exists between the image pairs. To overcome the spatial inconsistency between the features, an alignment unit, referred to as a gradient constraint unit, is proposed, which is shown in Fig. 4.

Most of the image fusion methods currently rely on training and evaluating airborne remote sensing images, from which the degradation image pairs are simultaneously simulated. As a result, the fused space-borne images often suffer from spatial inconsistency, where the edges of the ground objects become blurred and do not align well with the edges in the multispectral imagery. This artifact is closely related to the satellite platform motion and can be considered as anisotropic spread. To address this issue, image-gradient-based algorithms can be employed to constrain the spatial alignment of the image features and eliminate artifacts in the fused image.

To address this problem, a gradient-based feature alignment structure, termed as the gradient constraint unit, is proposed for image fusion. The image gradient indicates the direction of each pixel in the image, resembling a potential field. The spread effect occurs when inconsistent features with misaligned gradients are fused. Since the LR hyperspectral and HR multispectral images have separate residual estimations, simply concatenating the extracted features from them is insufficient. Therefore, a gradient constraint unit is constructed to learn a spatial constrain, to spatially align the feature maps from the HR multispectral images and LR hyperspectral images. Instead of directly estimating shrinkage parameters, which may result in a fixed size limitation of the input, a block for estimating shrinkage parameters has been proposed. Furthermore, the shrinkage parameter is estimated by the align part learned from the image gradient, which is obtained by the Sobel operator.

The grid sample unit, leveraging the directional characteristics of the image gradient, effectively shrinks the network based on the directionality. This constrains the unaligned spectral-spatial representation within the same network unit, aiming to suppress the image motion effect in the real image data and eliminate artifacts in the fused hyperspectral imagery.

E. Implementation Issues

Since the target image is always lacking in real remote sensing satellite image observations, obtaining simulated data by degraded images is introduced for the image training. Specifically, in order to avoid data insufficiency when training at the original resolution, a self-data-driven strategy is implemented by training on the degraded LR hyperspectral and HR multispectral images degraded from the original data. The LR hyperspectral image is simulated by spatial blurring and down-sampling from the original LR hyperspectral image, and the HR multispectral image is simulated by spatial blurring and down-sampling from the original HR multispectral image. The spatial blurring is simulated by a Gaussian kernel PSF. The image is then down-sampled to obtain a smaller size. Since spatial degradation is easy to approximate, the simulated data and original LR hyperspectral images can form training data pairs for training.

Similar to the construction of other simulation datasets, the proposed WHU-MHF dataset was constructed by spatially downscaling the hyperspectral imagery at a 30-m resolution and the multispectral imagery at a 10-m resolution, according to the Wald's protocol, to obtain simulated hyperspectral and multispectral images with resolutions of 90–30 m, respectively. The process can be expressed as follows:

$$\widetilde{\mathbf{I}} = \mathbf{I}^{*}\mathbf{G}^{*}\mathbf{P} \tag{12}$$

where the tensors $\mathbf{I}$ and $\widetilde{\mathbf{I}}$ denote the original image and the degraded image, respectively. According to the imaging model (12), when transferring through an optical system, the signal from the ground will be degraded and the imagery will turn out to be blurred. This process can be described

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as convolution by the PSF. Here, convolution in the spatial domain is applied by a Gaussian kernel filter $\mathbf{G}$ to achieve a similar blurring effect to the original image. As for the spatial down-sampling, average pooling is introduced to obtain a smaller image, which is described by the matrix $\mathbf{P}$. For the implementation of MH-FUNet, ten iterative fusion units are introduced.

In the proposed network, smooth $L1$ loss is introduced as the loss function of the network because this loss converges faster than mean squared error (MSE) loss, while also being robust to outliers [55]. This helps the network efficiently capture the deep fusion priors. The smooth $L1$ loss can be expressed as

$$\text{Loss}\left(\widehat{\mathbf{X}}, \mathbf{X}\right) = \sum_{i=1}^{n} \text{Smooth}_{L_1}\left(\widehat{\mathbf{X}}_i - \mathbf{X}_i\right) \tag{13}$$

$$\text{Smooth}_{L_1}(x) = \begin{cases} 0.5x^2 & \text{if } |x| < 1 \\ |x| - 0.5 & \text{otherwise.} \end{cases} \tag{14}$$

IV. Experiments

A. Experimental Settings

In the experiments, to demonstrate the efficiency of the proposed MH-FUNet framework, it was compared with several advanced multispectral and hyperspectral image fusion methods, that is, adaptive Gram-Schmidt (GSA) [11], HySure [56], Fusformer [57], DBIN [36], and MHF-Net [58]. For a fair comparison, all the parameters were the default settings or fine-tuned as suggested by the authors. In addition, the WHU-MHF image fusion dataset was introduced as a benchmark spaceborne multispectral and hyperspectral image fusion dataset, which was generated using paired multispectral and hyperspectral images from ZY-1-02D, including a simulated and a real subset. For the hardware, one RTX 3090 GPU was used for the training. The model was trained for 1000 epochs, the optimizer was the Adam optimizer, the parameter β was set to (0.5, 0.999), and the initial learning rate was set to 0.0001.

The experiments consisted of two parts: a simulation fusion experiment and a real-data fusion experiment. The initial motivation is satellite image fusion; however, the absence of reliable reference real images necessitated the inclusion of the airborne dataset. Exploiting the high spatial and spectral resolution of the airborne dataset, quantitative accuracy evaluations could be performed convincingly. In the case of the spaceborne dataset, a visual comparison of the proposed network with the reference image was carried out to illustrate its effectiveness in real-world image fusion tasks. To ensure the accuracy of the image spectrum and prevent spectral aliasing caused by reduced spectral resolution, typical features were first identified in the uniform area of the original hyperspectral image based on visual observation. The coordinates of these features were recorded. Following fusion, considering a threefold resolution difference between image pairs, the recorded coordinates were scaled by three times. Subsequently, these scaled coordinates were utilized as an index to select the corresponding spectral curve of the fused object for visual display. The simulation experiment was conducted on the airborne hyperspectral Chikusei dataset. The original image was cropped to 2000×2000 pixels to avoid the black edges of the original image and then separated into 64 subimages. Due to the high quality of the image, these image patches could be directly used as the ground truth (GT) for the quantitative evaluation. The experiment on real data was conducted on the WHU-MHF dataset, and the details are different from the simulation experiment. For the CS and variational-based methods, no training process is needed so that the fusion process is directly conducted at the original resolution. As for the deep-learning based-methods, the network was first trained on the down-sampled 90–30 m image pairs, which are generated from the original ones. Then inference was conducted on the images with the original 30–10 m resolution.

Four common metrics are employed in this article for the quantitative assessment. The mean value of the peak SNR (PSNR) can be denoted as

$$\text{PSNR} = \frac{1}{S}\sum_{k=1}^{S}\left(10\log_{10}\frac{\text{MN}}{\sum_{i=1}^{M}\sum_{j=1}^{N}\left(\widehat{\mathbf{X}}_{i,j,k} - \mathbf{X}_{i,j,k}\right)^2}\right). \tag{15}$$

The mean structural similarity (SSIM) can be denoted as

$$\text{SSIM} = \frac{1}{S}\sum_{k=1}^{S}\left(\frac{(2\mu_{\mathbf{X}}\mu_{\mathbf{X}} + C_1)(2\sigma_{\mathbf{X}\widehat{\mathbf{X}}} + C_2)}{(\mu_{\mathbf{X}}^2 + \mu_{\widehat{\mathbf{X}}}^2 + C_1)(\sigma_{\mathbf{X}}^2 + \sigma_{\widehat{\mathbf{X}}}^2 + C_2)}\right) \tag{16}$$

where μ and σ, respectively, represent the mean and variance of the image and C represent constants.

The spectral angle mapper (SAM) can be denoted as

$$\text{SAM} = \frac{1}{\text{MN}}\sum_{i=1}^{\text{MN}}\arccos\frac{\mathbf{X}_i^T\hat{\mathbf{X}}_i}{\sqrt{\mathbf{X}_i^T\mathbf{X}_i}\sqrt{\hat{\mathbf{X}}_i^T\hat{\mathbf{X}}_i}}. \tag{17}$$

The mean relative dimensionless global error in synthesis (ERGAS) can be denoted as

$$\text{ERGAS} = \frac{100}{\lambda}\sqrt{\frac{1}{S}\sum_{k=1}^{S}\left[\frac{\sum_{i=1}^{M}\sum_{j=1}^{N}\left(\widehat{\mathbf{X}}_{i,j,k} - \mathbf{X}_{i,j,k}\right)^2}{\text{MN}\mu_{\mathbf{X}_k}^2}\right]}. \tag{18}$$

The smaller the ERGAS value, the better the fusion result.

To evaluate the performance of the algorithms on the simulated dataset, the mean values of the evaluation matrices, denoted as MPSNR, MSSIM, MERGAS, and MSAM, are utilized to assess the fusion quality. For experiments on real datasets, the image fusion results are visually displayed. Specially, to guarantee the precision of the image spectrum and prevent spectral aliasing resulting from reduced spectral resolution, a systematic approach was employed. Initial steps involved the identification of representative features within the uniform regions of the original hyperspectral image through visual observation, with subsequent recording of the corresponding image coordinates. Following the fusion process, given a threefold disparity in resolution between image pairs, a scaling factor of three was applied to the recorded coordinates. These scaled coordinates were then utilized as indices to retrieve the associated spectral curve of the fused object, facilitating its visual representation.

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TABLE I
DETAILS OF THE PROPOSED WH-MHF DATASET

Index of the scene	LR hyperspectral image size	HR multispectral image size	Images Patches		
			Train	Evaluation	Total
1	1628×1556×148	4884×4668×8	76	80	156
2	1848×776×148	5544×2298×8	49	41	90
3	1326×739×148	3978×2217×8	29	37	66
4	1471×1963×148	4413×5889×8	99	93	192
5	657×1019×148	1971×3057×8	0	40	40
6	869×962×148	1290×1738×8	56	0	56
7	1290×1738×148	3870×5214×8	67	73	140

B. Experimental Datasets

1) Airborne Chikusei Dataset for the Simulation Experiment: The Chikusei dataset consists of hyperspectral images captured in Tsukishi Ibaraki, Japan [59]. The data was acquired using the Headwall Hyperspec-VNIR-C imaging sensor, covering agricultural and urban regions in Chikusei, Ibaraki, Japan. The hyperspectral dataset comprises 128 bands covering the spectral range from 363 to 1018 nm. The scene has dimensions of 2517 × 2335 pixels, with a ground sampling distance of 2.5 m.

For the experiments, the entire image was subdivided into 64 subimages, each measuring 250 × 250 pixels, and these subimages were used as reference data. Subsequently, the subimages were further degraded to generate LR multispectral images and HR hyperspectral images for evaluation purposes.

2) Proposed Spaceborne WHU-MHF Dataset for the Real-Data Experiment: The real-data experiment consisted of LR hyperspectral images and HR multispectral images with the ground sample distance of 30–10 m, respectively. The whole WHU-MHF dataset consists of six pairs of LR hyperspectral and HR multispectral images. In the experiment, the hyperspectral images were divided into 40 × 40 overlapping patches, and the multispectral images were divided into 120 × 120 overlapping patches, with a total of 740 image pairs, 50% of which were randomly selected for the training dataset and the remaining for the testing. To eliminate the possibility of data leakage, the whole No. 5 image is used as the test part, on which the visualization was conducted. More details of the proposed WHU-MHF dataset are provided in Table I.

C. Simulated Experiment With the Chikusei Dataset

The visual and quantitative evaluations of the fusion results are provided in Figs. 5 and 6, and Table II, respectively. From the results, it can be seen that GSA cannot achieve good fusion results, as a result of the loss of image details, which is also indicated in the low quantitative results. The fused image from HySure contains severe artifacts and a lot of spectral distortion, leading to a high MERGAS value in the fusion result. The result of MHF-Net also contains severe artifacts. This method is based on a spatial image fusion strategy. When image distortion appears between the images, the algorithms cannot capture the correct spatial information. GSA can achieve great fusion results, but the spatial spread effect still exists. DBIN and MH-FUNet achieve a great performance in both the spatial detail and spectral signal restoration. Due to the attention unit, the proposed MH-FUNet framework can obtain both a higher image fusion quality and better spectral fidelity.

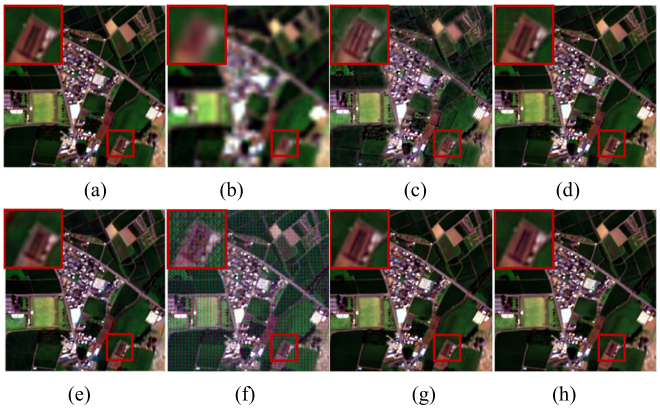


Fig. 5. Visual comparison for the fusion results on the Chikusei dataset by true-color synthesis (red: 60, green: 40, blue: 20). (a) GT. (b) LR image. (c) GSA. (d) HySure. (e) Fusformer. (f) MHF-Net. (g) DBIN. (h) MH-FUNet.

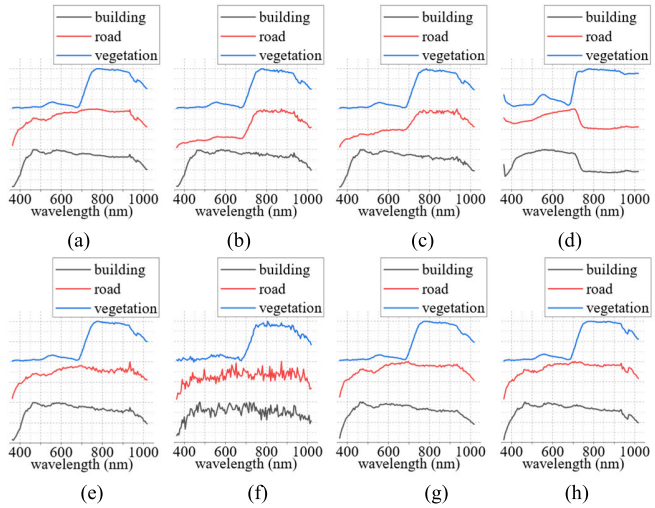


Fig. 6. Spectral comparison of the spectra for homogeneous regions in the fused images. (a) GT. (b) LR image. (c) GSA. (d) HySure. (e) Fusformer. (f) MHF-Net. (g) DBIN. (h) MH-FUNet.

TABLE II
QUANTITATIVE EVALUATION OF THE FUSION RESULTS FOR THE CHIKUSEI DATASET

Method	MPSNR	MSSIM	MSAM	MERGAS
GSA	33.24	0.89	9.51	5.66
HySure	10.68	0.43	9.91	74.41
Fusformer	44.10	0.99	**2.22**	1.30
MHF-Net	34.99	0.86	6.77	4.46
DBIN	43.70	0.99	3.40	1.70
MH-FUNet	**44.51**	**0.99**	2.58	**1.28**

D. Real-Data Experiment With the WHU-MHF Dataset

To demonstrate the effectiveness of the proposed method in spaceborne image fusion, an experiment on the WHU-MHF dataset was conducted. The results are shown in Figs. 7 and 8.

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Fig. 7. Visual comparison of the fusion results obtained with the WH-MHF dataset by true-color synthesis (red: 29, green: 19, blue: 10). (a) HR multispectral image. (b) GSA. (c) HySure. (d) Fusformer. (e) MHF-Net. (f) DBIN. (g) MH-FUNet.

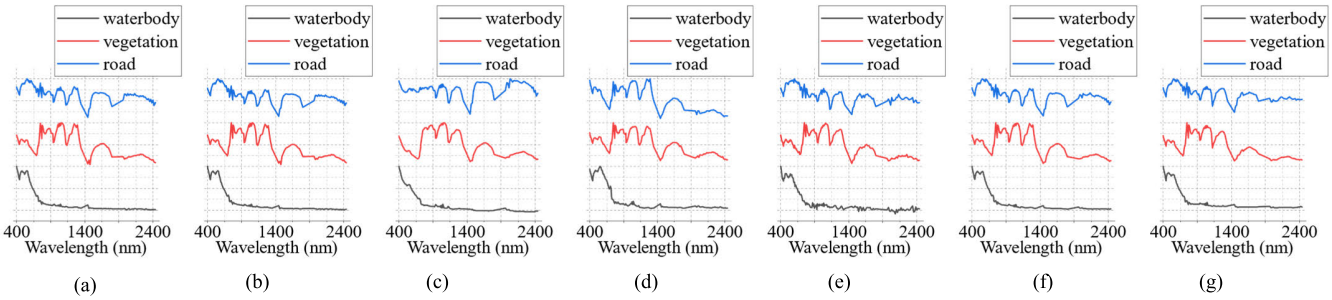


Fig. 8. Spectral comparison of the fusion results obtained with the WH-MHF dataset by spectral visualization. (a) LR hyperspectral image. (b) GSA. (c) HySure. (d) Fusformer. (e) MHF-Net. (f) DBIN. (g) MH-FUNet.

GSA fails to capture the correct spatial component of the fusion images. That is because the spread effect has corrupted the spatial consistency between the original LR hyperspectral image pairs and HR multispectral ones. Due to the fusion of hyperspectral images using spectral super-resolution mode in HySure, sharper edge details better edges can be obtained as a result. The spectral restoration is good, but the spatial features are not well learned, and the edge of the fused image is relatively blurred. For MHF-Net, the restored edge is sharper, but there are serious grid artifacts, and the spectral are biased, which indicates that gradient-based image prior, which is utilized in MHF-Net, requires high spatial consistency between image pairs. For DBIN, there is a small number of artifacts, and the restoration is generally good, but the features still spread out at the edges. The proposed MH-FUNet framework results in no artifacts or blurring and obtains the most similar result to the HR multispectral image.

E. Ablation Study

To illustrate the effectiveness of the proposed architecture, we conducted further ablation experiments on both Chikusei dataset and WHU-MHF dataset. The visual and quantitative evaluations of the fusion results are provided in Figs. 9 and 10, and Table III, respectively.

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TABLE III
QUANTITATIVE EVALUATION OF THE ABLATION STUDY FOR THE CHIKUSEI DATASET

Method	MPSNR	MSSIM	MSAM	MERGAS
w/o proposed fusion units	40.14	0.97	3.78	2.39
w/o gradient constraint units	41.88	0.98	2.95	1.73
MH-FUNet	**44.75**	**0.99**	**2.55**	**1.28**

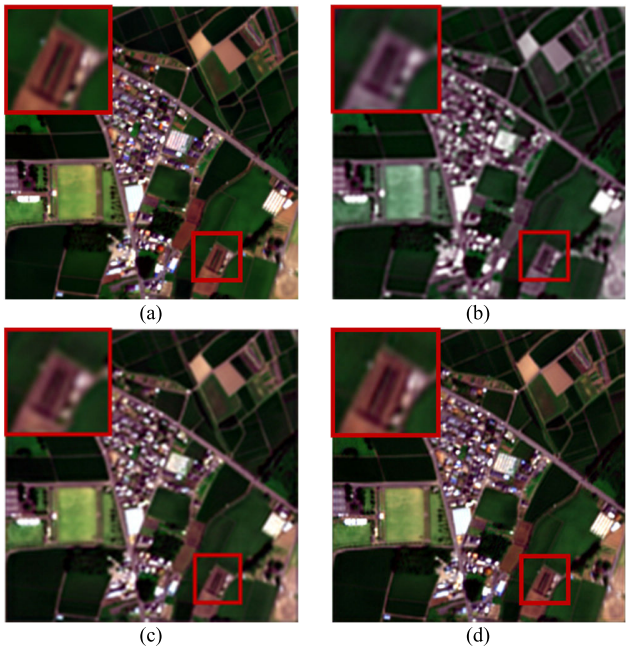


Fig. 9. Visual evaluation for the ablation study on Chikusei dataset by true-color synthesis (red: 60, green: 40, blue: 20). (a) HR hyperspectral. The results in column (b) were from MH-FUNet where gradient constraint unit were kept while the multiscale fusion unit were replaced with convolutional blocks. The results in column (c) came from MH-FUNet with only gradient constraint unit removed. (d) MH-FUNet.

TABLE IV
QUANTITATIVE EVALUATION OF THE TIME AND MEMORY ON WHU-MHF DATASET

Method	Time(s)	Parameters(MB)
GSA	825.19 (CPU)	-
HySure	2151.24 (CPU)	-
Fusformer	**2.46**	**1.98**
MHF-Net	3.14	27.29
DBIN	3.24	72.84
MH-FUNet	29.06	271.89

The proposed fusion units in MH-FUNet were initially removed to establish a comparison basis, as indicated in Figs. 9(b) and 10(b). In the experimental outcomes, the contour of the terrain exhibited slight disturbance, resulting in a certain degree of blurring. Concerning the image color, reflective of the spectral accuracy, a noticeable shift in the color tone was observed, highlighting the effectiveness of the proposed

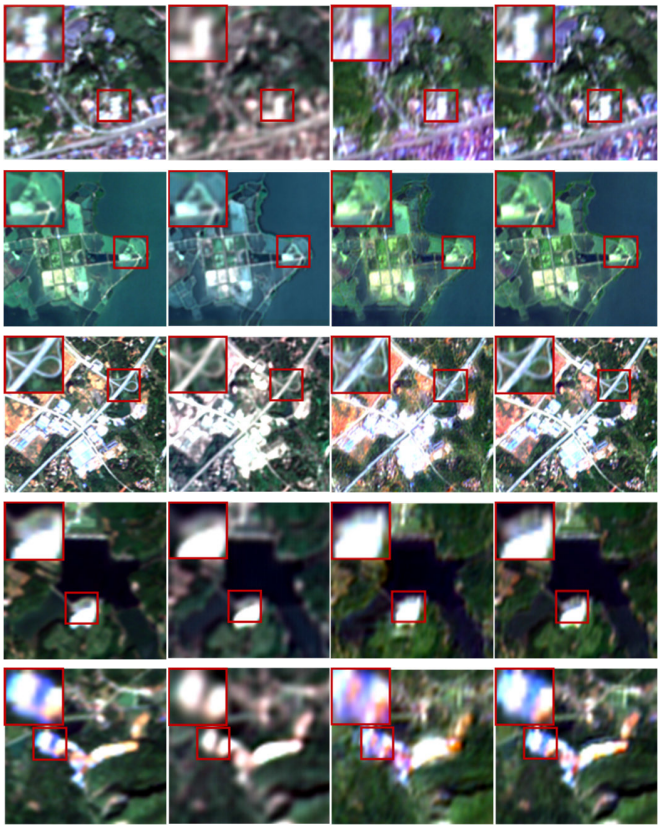


Fig. 10. Visual evaluation for the ablation study by true-color synthesis (red: 29, green: 19, blue: 10). (a) HR multispectral. The results in column (b) were from MH-FUNet where gradient constraint unit were kept while the multiscale fusion unit were replaced with convolutional blocks. The results in column (c) came from MH-FUNet with only gradient constraint unit removed. (d) MH-FUNet.

fusion module in preserving image spectral fidelity. Moreover, to illustrate the impact of the proposed gradient constraint unit on addressing image spatial inconsistencies, a version of MH-FUNet without the gradient constraint unit was employed for comparison, corresponding to Figs. 9(c) and 10(c). The results for MH-FUNet without any proposed units appeared to be the most blurred. In contrast, the comparison with MH-FUNet lacking the gradient constraint units revealed that the employed gradient constraint unit within the network effectively aligns spatial features and mitigates the spread effect in the fusion results for further comparison of the efficiency of the methods. Evaluation on time and memory are provided in Table IV. The inference time on the dataset is used as a reference for statistical analysis, with *batch size* set to 1 during inference. For memory usage, the size of network parameters is used for comparison.

V. CONCLUSION

In this article, in view of the scale-dependence with deep learning methods and the difficulty of accurate representation of image spatial-spectral features, a deep self-supervised unrolling network (MH-FUNet) has been proposed for multispectral and hyperspectral image fusion based on multiscale spatial-spectral attention. The multiscale image fusion learning is carried out through the unrolling technique and the fusion mapping under different scales is mined. The spectral and

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spatial detail information of the imagery is characterized and stored by introducing the spatial attention mechanism for the multispectral imagery and the channel attention mechanism for the hyperspectral imagery, so as to supplement the spatial information of the fused image obtained by the multiscale learning unit. The gradient constraint unit is used to achieve spatial consistency constraints for sampling different-scale features and suppressing the spread effect in the fused image.

Experiments on a simulated dataset showed that the MH-FUNet model proposed in this article can better integrate multispectral and hyperspectral images and can maintain better spectral and spatial details. The real-data experiment further proved that MH-FUNet has scale-independence and a strong generalization ability. Compared with the other fusion methods, the model trained on simulated data can be migrated well to real data and performs more stably. To sum up, the MH-FUNet model proposed in this article is accurate, efficient, and practical and could serve hyperspectral satellite data production and related application industries, such as ZY-1-02D. Despite the good performance on the proposed dataset, the trained model could not be used on another dataset. How to dynamically adapt to different channel numbers of data between different sensors and their data characteristics still needs further research.

VI. Data Availability

The dataset and code used in this study are now openly available for use on request by other researchers. Interested parties are encouraged to contact us via e-mail to obtain a copy of the dataset and the code.

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